M101 · Unit 9 · Quick reference

Choosing an integration method
Version 1.0 · 23 October 2026 · 1 / 2

Recall **Elementary pairs**	$\int x^p dx = \frac{x^{p+1}}{p+1} + C$, $p \neq -1$; valid real interval. $\int dx / x = \ln \lvert x \rvert + C$, interval excludes 0.
Recall **Exponential and trig**	$\int e^x dx = e^x + C$; $\int \cos x dx = \sin x + C$; $\int \sin x dx = -\cos x + C$. Radians.
Recall **Linear input**	If $F' = f$, $a \neq 0$: $\int f(ax+b)dx = \frac{1}{a}F(ax+b) + C$.
Understand **Substitution**	$u = g(x)$, $du = g'(x)dx$; rewrite every factor. Check by the chain rule.
Recall **Changed limits**	$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$. Preserve endpoint order.
Recall **Parts**	$\int u\,dv = uv - \int v\,du$. Choose an easier remaining integral; finish with C.

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Version 1.0 · 23 October 2026 · 2 / 2

Understand **Definite parts**	$\int_a^b uv'dx = [uv]_a^b - \int_a^b u'vdx$. Same variable and bounds.
Consult **Conditions**	FTC: continuous integrand on closed interval. Substitution: g continuously differentiable; f continuous over its image. Definite parts: u,v continuously differentiable.
Understand **Verification**	Antiderivative: $F' = f$ on interval. Definite result: $F(b) - F(a)$. Initial value determines C. Numerical samples supplement identities.
Consult **Method choice**	Simplify → elementary pair → inner derivative match → useful parts split. Reinspect after each transformation. Domain, signs and units remain part of the calculation.

Consult · Python

```
def F(x):
    return x**3

def change(F, a, b):
    return F(b) - F(a)

print(change(F, 0.0, 1.0))  # 1.0
print(change(F, 1.0, 0.0))  # -1.0
```

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